Théo Gigant

Research Scientist

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HuggingFace: gigant

Education

2022–2025 **Université Paris-Saclay / Centrale Supélec - L2S / LISN**,
Signal and Image Processing, PhD.
Designing and evaluating methods for automatic summarization of multimodal presentations based on LLMs and VLMs. Under the supervision of Frederic Dufaux and Camille Guinaudeau.

2018–2021 **Institut Polytechnique de Paris - Télécom SudParis**,
Statistical Modeling, Master of Science.
Applied Mathematics, Machine Learning, Deep Learning, Computer Vision

Professional Experience

Aug 2025–Ongoing **Research Scientist**, Nous Research, New York City.
Multimodal and byte language models.

Jul 2022–Jul 2025 **PhD Researcher**, Centrale Supélec, Paris.
Conducting research on summarization of multimodal presentations using multimodal deep learning methods.

Jul 2022–Dec 2024 **PhD Researcher**, JustAI, Paris.
Building machine learning pipelines, for applications such as presentation summarization, keyframes extraction, image segmentation, tracking & recommender systems. Teaching deep learning.

Jan 2022–Jun 2022 **Machine Learning Engineer**, Freelance, Paris.
Developing image segmentation & detection models, teaching deep learning, contributing to open source projects.

Jul 2021–Dec 2021 **Computer Vision Research Intern**, Huawei R&D Center, Helsinki.
Developing multimodal models for long video understanding, video retrieval & image generation.

Feb 2021–Jun 2021 **Machine Learning Engineer Intern**, JustAI, Paris.
Proofs of concept with language models, developing data processing and data visualization pipelines.

Jul 2019–Aug 2019 **Machine Learning Research Intern**, Télécom SudParis, Evry.
Developing image classification datasets & models.

Projects

2025 **Multimodal Summarization Toolkit [Repo]**, Slide extraction, transcription, visualization and semantic search for slide-based multimodal presentations..

2022-Ongoing **Deep Learning blog posts [Blog]**, Insights, reflections and opinions on VLMs and LLMs.

Apr 2022–May 2022 **BigBio BioMedical Dataset project [Org]**, *BigScience*, paper published at NeurIPS 2022.

Apr 2022–May 2022 **WikiArt Diffusion Mini [Project]**, *HuggingFace & Paperspace*, HugGAN Challenge.

2022 **ASR Models for Romanian: Wav2Vec2 and Whisper**, *HuggingFace*, Speech Recognition Challenges (Winner).

July 2021 **T5-VAE [Project]**, *Google & HuggingFace*, JAX/Flax Community Week.

First-Author Publications

Apr 2025 **Summarization of Multimodal Presentations with Vision-Language Models: Study of the Effect of Modalities and Structure**, preprint.
Fine-grained evaluation of VLMs to devise optimal input settings for summarization of multimodal presentations.

Nov 2024 **Mitigating the Impact of Reference Quality on Evaluation of Summarization Systems with Reference-Free Metrics**, *EMNLP*.
Introducing a new metric for reference-free evaluation of the relevance of summaries. Evaluating the robustness of metrics with respect to the quality of references to develop a robust hybrid evaluation strategy.

Sep 2023 **TIB: A Dataset for Abstractive Summarization of Long Multimodal Videoconference Records**, *CBMI* .
Introducing a dataset of 9k multimodal presentations with a benchmark of fine-tuned state-of-the-art long document summarization models.

Skills and Research Interests

Languages Python (JAX/Flax, PyTorch, transformers), Bash, SQL, Java

Research Interests Large-scale model training (trained on Jean-Zay HPC cluster), Vision Language Models, Large Language Models, Evaluation, Multimodality, Document Understanding

Spoken Languages French (native), English (Full professional proficiency : 990/990 TOEIC), Spanish (Notions)